DSE Bullet Series – System Software (Solutions 04)

Name: _____ ( ) Class:____ Date: _____

1. Past Questions (2014–2022) #System or application — don't mix them up

Year	Past Question	Focus / Marks
2014	Freeware vs open source software	Both free; source code disclosed or not
2015	Shareware vs open source software	Trial limits; source code open and editable
2016	Standalone software vs cloud service	Buy + install vs subscribe + browser
2018	Two differences: Linux / macOS / Windows	Interface, compatibility, licence cost
2020	System software vs application software	Manage hardware/platform vs user tasks
2021	Why files transfer between different OS?	Common file formats and network protocols
2022	Two OS functions besides file management	Memory, process, I/O, user interface

2. Key Notes #One manages, one helps

System software manages hardware and provides a platform: operating system, device drivers and utility software. Application software helps users complete tasks: word processing, spreadsheets, browsers and accounting.

OS functions: memory, process, input/output and file management; user interface; error handling.

Device drivers let the OS communicate with specific hardware; utility software handles antivirus, backup, disk cleanup, compression and system monitoring.

Computer system types: batch processing (monthly statements); online interactive (immediate response); real-time (response within a time limit); parallel (multiple processors in one system); distributed (multiple computers at different locations share work over a network); virtualisation.
2016 DSE Q17 (Answer: A real-time system): counters at different locations are only terminals, not separate computers; seat availability must be confirmed and updated immediately. Memory tip: distributed = multiple computers across locations; parallel = multiple processors in one system.

Licensing: commercial software needs a licence; freeware does not disclose source code; open source software discloses and allows modification of source code; shareware has trial or feature limits.

Cloud vs local: cloud software needs no installation, works across devices and updates automatically; it requires a network and has higher long-term cost.
*Common mistakes: assuming ‘freeware = open source software’; confusing system and application software; writing only ‘convenient’ or ‘easy to use’; comparing without a common criterion.

3. Similar Questions #Same angle, different software

1. The school uses a cloud word-processing service, not standalone software.

(a) State **two** benefits of switching to a cloud service.

(b) State one risk of using a cloud service.

2. The technician adds application software after installing the operating system.

(a) State **two** types of application software and explain their purposes.

(b) Explain one difference between application software and system software.

3. The school is considering replacing commercial software with open source software.

(a) State **two** benefits of using open source software.

(b) Explain why 'freeware' is not necessarily the same as 'open source software'.